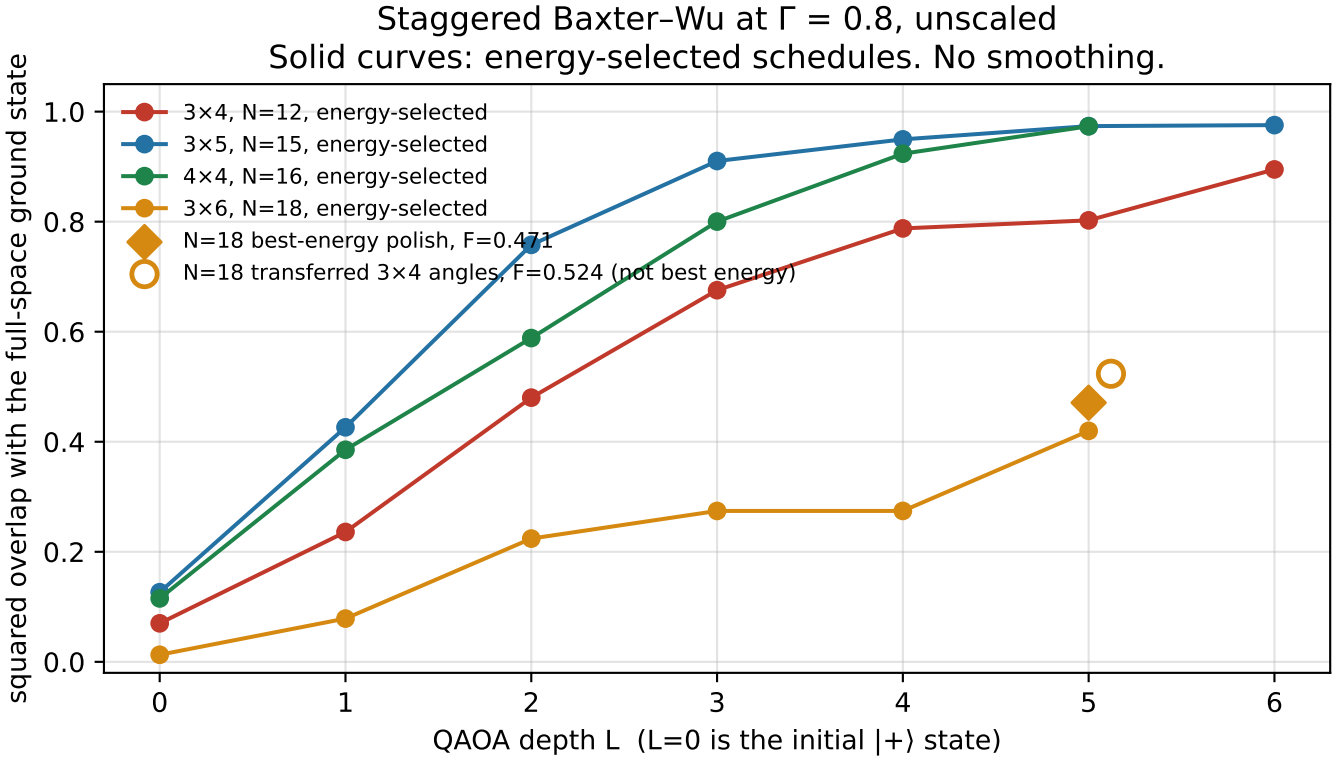

N=12 is 3×4 (branch A). 4×3 is the same N=12 Hamiltonian and is not drawn as another size.
N=18 solid endpoint is the blind 8-restart search (F=0.420). The diamond is a lower energy; the open marker is a higher overlap.